

Bambusa reports positive phase 1 results for BBT001 (anti-IL-4Rα/IL-31 bispecific) in AtD



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Published 7/29/2026

Summary

Bambusa reported preliminary proof-of-concept (POC) data from the ongoing global phase 1 [trial](#) of BBT001, a half-life-extended bispecific antibody targeting IL-4Rα and IL-31, in bio-naïve patients with moderate-to-severe atopic dermatitis (AtD). In a small cohort (n=17; 12 active, 5 placebo), BBT001 450 mg IV Q2W for 4 weeks produced highly statistically significant placebo-adjusted EASI reductions beginning at week 1 (−35.56%, p=0.0012), deepening through week 6 (−78.95%, p<0.0001). Rapid itch relief was observed from day 1, no conjunctivitis was reported, and the long half-life supports potential Q3M maintenance dosing. Phase 2b planned; additional IV data (12-week AtD bio-naïve/bio-experienced, 14-week SC) expected 1H'27

Why it Matters: BBT001 is the only IL-4Rα/IL-31 bispecific antibody in active clinical development after J&J terminated its IL-4Rα/IL-31 bispecific (JNJ-64737283) due to not meeting the “high bar” of efficacy in a phase 2b study for atopic dermatitis. The positive efficacy is encouraging but not surprising since IL-4 and IL-31 are validated targets. However, it is too early to claim BBT001 is the “best in disease” based on data from 17 patients. A dupilumab-like future phase 2b study like the one of J&J will be imperative to show the true potential of BBT001.

Key Highlights

- BBT001 is a novel tetravalent “2+2” bispecific antibody engineered to simultaneously target the IL-4/IL-13 and IL-31 pathways, featuring FcRn-mediated recycling and Fc modifications to extend its half-life
- Efficacy results: Data cutoff: June 8, 2026
 - EASI efficacy (placebo-adjusted % change; observed between-group differences, p-values from MMRM): wk1 −35.56% (p=0.0012), wk2 −51.11% (p<0.0001), wk4 −63.46% (p<0.0001), wk6 −78.95% (p<0.0001). EASI reduction continued to deepen through wk12 and was sustained for patients followed
 - Additional exploratory endpoints (placebo-adjusted, observed between-group differences): EASI-50 — wk1 25%, wk2 75%, wk4 82%, wk6 82%. EASI-75 — wk1 0%, wk2 17%, wk4 45%, wk6 64%. PP-NRS % change — wk1 −29%, wk2 −35%, wk4 −41%, wk6 −43%. Pruritus improvement began as early as day 1 and was sustained for 8 weeks after the last dose
- Biomarkers: Early and substantial reductions in TARC and IgE, sustained 8 weeks post-last dose — confirming durable Type 2 pathway suppression
- Safety: BBT001 was well tolerated. No conjunctivitis reported (consistent with prior healthy volunteer findings). Low immunogenicity, no ADA titers, no apparent neutralizing activity
- PK: Extended half-life in AtD patients consistent with healthy volunteer data, supporting potential maintenance dosing as infrequent every 3 months (Q3M)
- Key upcoming catalysts: Full results at upcoming medical conference (TBD). 1H 2027 — topline 12-week IV data in bio-naïve and bio-experienced AtD; 14-week IV data in CSU. Phase 2b initiation in moderate-to-severe AtD evaluating SC maintenance dosing up to Q3M
- J&J terminated its IL-4R/IL-31 after an interim analysis of a phase 2b with dupi as comparator. The company said the molecule “did not meet the high-bar efficacy we established for advancing our clinical development programs for atopic dermatitis.” ([link to VITAL post](#))
- Read the data in more detail ([link](#)). Detailed cross-trial indirect comparison data are presented in the chart below

Details

- Trial design:** This study is part of a global, randomized, double-blind, placebo-controlled Phase 1 clinical trial (NCT06808477) evaluating IV and SC formulations of BBT001 in healthy volunteers and AtD patients. The AtD portion enrolled bio-naïve patients with moderate-to-severe disease (no prior biologics targeting the same pathways as BBT001, no prior JAKi), randomized 2:1 to BBT001 450mg IV Q2W or placebo for a 4-week treatment period at sites in New Zealand and the United States.
- Baseline Characteristics:**

Parameter	BBT001 (n=12)	Placebo (n=5)
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EASI, mean (SD)	34.64 (11.38)	29.06 (14.0)
BSA %, mean (SD)	62.58 (19.87)	47.60 (25.1)
Weekly PP-NRS, mean (SD)	7.34 (0.87)	6.33 (1.55)
vIGA, mean (SD)	3.42 (0.51)	3.20 (0.44)

- Efficacy — EASI reduction (placebo-adjusted): Analysis: Observed between-group differences; p-values from MMRM

Visit	Placebo-Adj. Change (%)	95% CI	
Week 1	–35.56	(–50.96, –13.59)	
Week 2	–61.10	(–81.24, –43.87)	
Week 4	–63.46	(–80.75, –43.27)	
Week 6	–78.95	(–94.62, –57.14)	

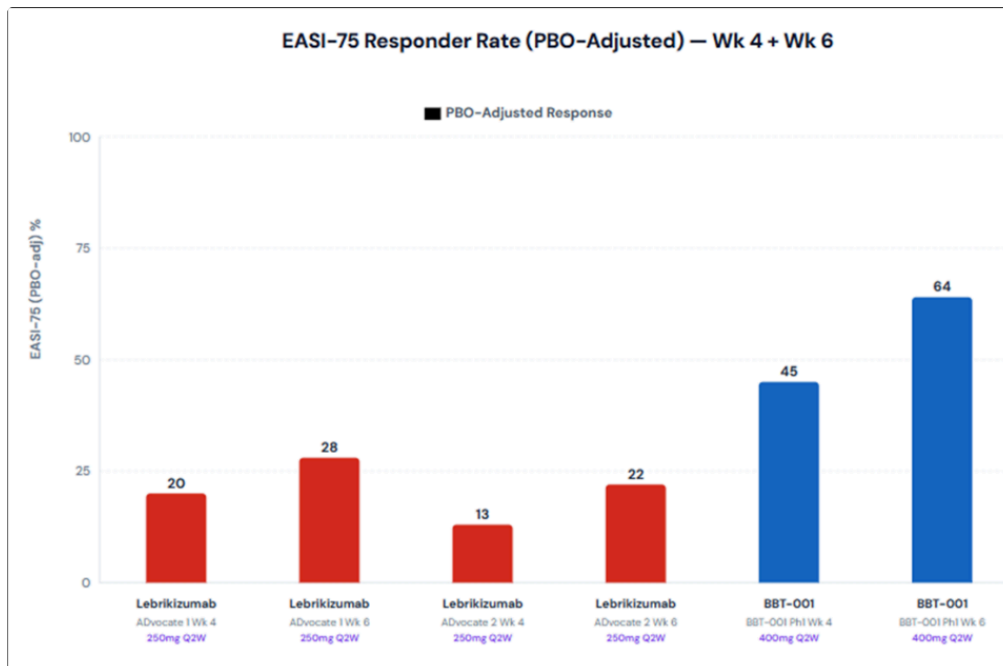
- Exploratory endpoints (placebo-adjusted)

Visit	EASI-50 (%)	EASI-75 (%)	PP-NRS Change (%)
Week 1	25	0	–29
Week 2	75	17	–35
Week 4	91	45	–41
Week 6	82	64	–43

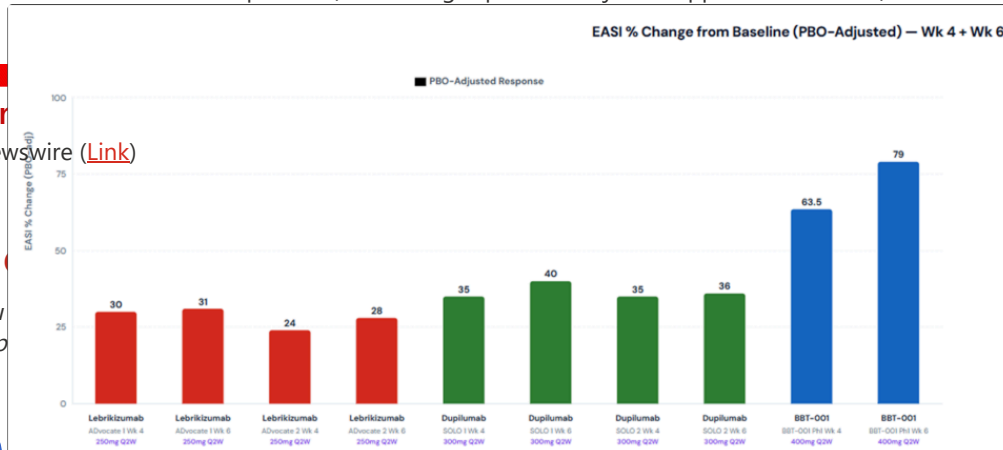
- Biomarkers: BBT001 produced early and substantial reductions in TARC and IgE that were sustained for 8 weeks after the last dose demonstrating durable inhibition of Type 2 inflammation.
- Safety: BBT001 was well tolerated. No cases of conjunctivitis were reported, consistent with prior findings in healthy volunteers.
- Immunogenicity: Low incidence of treatment-emergent anti-drug antibodies, with low titers and no apparent evidence of neutralization consistent with prior healthy volunteer findings.
- PK: BBT001 demonstrated an extended half-life in AtD patients, consistent with prior observations in healthy volunteers, supporting potential for maintenance dosing as infrequently as once every 3 months

Observations/Comments:

- BBT001 is a novel tetravalent “2+2” bispecific antibody engineered to simultaneously target the IL-4/IL-13 and IL-31 pathways, featuring modifications to extend its half-life
- Company pipeline: BBT002 (IL-4Rα/IL-5 — asthma, COPD, CRSwNP; Phase 1b/2a); BBT003 (GI, preclinical); BBT004 (autoimmune, preclinical)
- Data caveats — interpret with caution:
 - Very small N (17): Only 5 placebo patients
 - Baseline severity imbalance: Active arm entered with higher disease burden (EASI 34.64 vs 29.06; BSA 62.58% vs 47.60%). Regression to the mean could inflate observed treatment effects. The 95% CIs are wide accordingly
 - IV formulation only: 450 mg IV Q2W is not a commercial-stage dosing regimen. The Q3M maintenance dosing claim is based on extrapolation, not clinical demonstration at that dosing interval. SC PK/efficacy — critical for commercial viability — remain to be seen
 - Bio-naïve only: No bio-experienced patients in this cohort. Real-world AtD populations increasingly include patients with prior biologic exposure. Bio-experienced IV data expected 1H 2027
 - Exploratory endpoints: All efficacy endpoints are exploratory; the primary endpoint is safety/tolerability. Not powered for efficacy conclusions.
- Indirect cross trial comparison (EASI-75- placebo-adjusted approximate values)



- Indirect cross trial comparison (EASI change- placebo-adjusted approximate values)



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